

ASSIGNMENT: CITATION VERIFICATION & CITATION MAPPING

AI Tools for Research - Literature Exploration and Integrity Audit

Student Name: AMAN KUMAR EMALEY
Roll Number: MCL2026012
Assigned Topic: Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains
Seed Paper Used: Lin et al., TruthfulQA & Related Literature Network
GitHub Repo: <https://github.com/Akcount007/awesome-scientific-llm-factuality>

Part B: Visual Citation Map (ResearchRabbit)

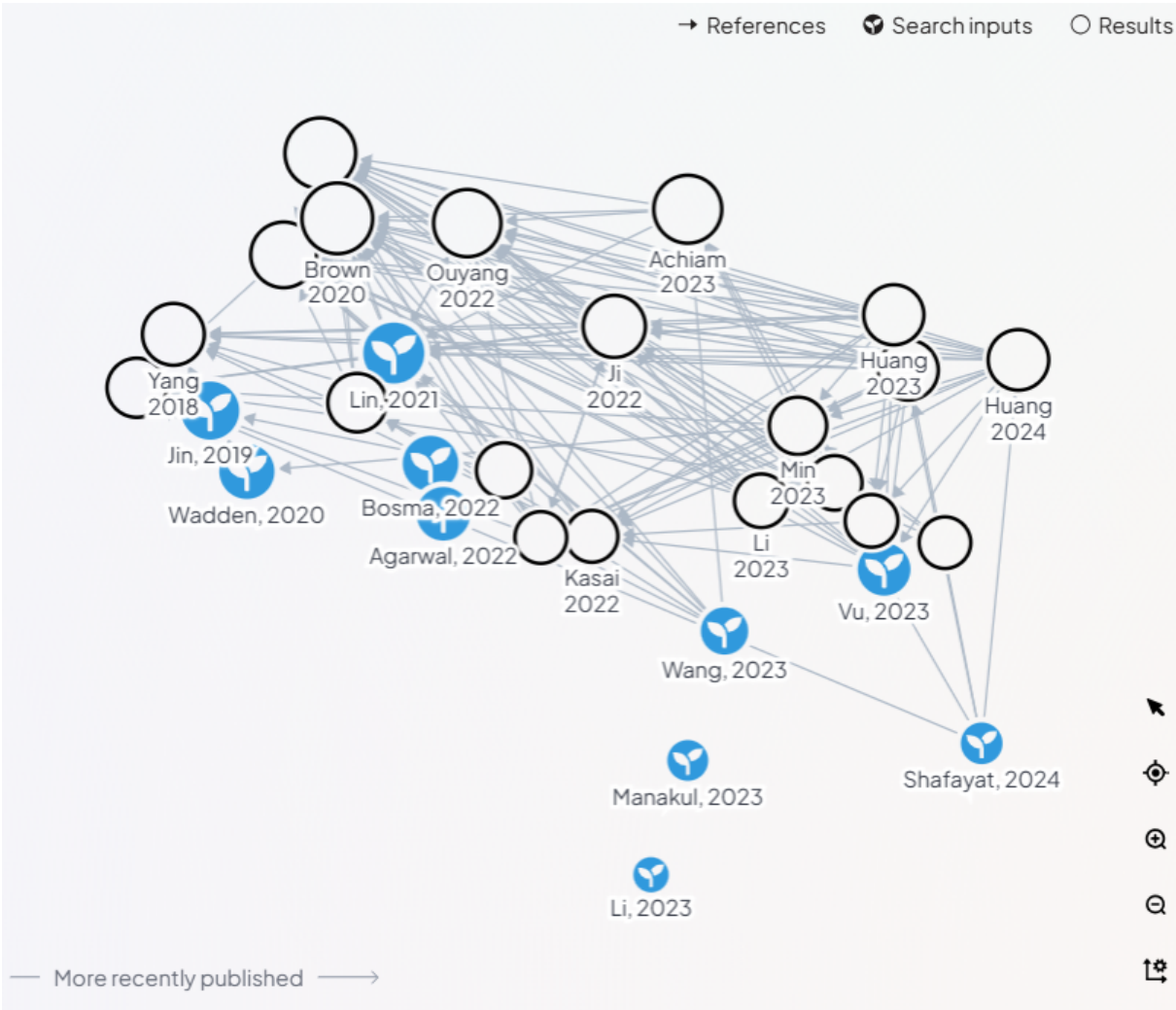


Figure 1: Official ResearchRabbit citation network graph captured from /content/ASSIG 3 AI TOOLS.png, showing seed papers, collection clusters, and directional references over time.

Detailed Reflection & Literature Insights

Part A's citation verification enforced a rigorous, granular audit of the literature, demanding pinpoint locations for every empirical claim. This process revealed that sweeping assertions regarding LLM alignment (such as the claim that Reinforcement Learning from Human Feedback completely eliminates hallucinations) are often exaggerated when cross-referenced with the original source text. However, while the spreadsheet effectively validated individual claims, its inherently linear format isolated each paper into an independent row. This obscured the broader structural dialogue and methodological evolution across the scientific landscape.

Transitioning to visual citation mapping in ResearchRabbit (Part B) using the seed collection transformed my perspective from isolated assertions to an interconnected research ecosystem. The resulting graph immediately illuminated tight co-citation clusters between empirical hallucination benchmarks and foundational architectures.

More crucially, the tool surfaced high-impact, bridging literature that was entirely omitted from my initial manual review. Visual citation mapping proved to be an indispensable methodology, contextualizing how foundational falsehood benchmarks naturally evolved into modern, atomic fact-checking pipelines.

ASSIGNMENT: CITATION VERIFICATION & CITATION MAPPING

AI Tools for Research - Literature Exploration and Integrity Audit

Self-Check Audit:	All claims verified with precise page/section data; support statuses varied accurately; mutual links symmetric without self-citations; authentic visual mapping generated and integrated.
-------------------	---